Conceptual question · Unit 3 · 2 points · 15 min · Conceptual / fast

C06 — Particle on a moving sphere: degrees of freedom, constraint type and the work of a smooth reaction

*A rigid sphere of radius R has its centre C on the vertical axis Oz of the inertial frame $S_0 = \{O;\mathbf{i},\mathbf{j},\mathbf{k}\}$ ($\mathbf{k}$ upward, gravity $-g\,\mathbf{k}$). A motor moves the sphere without rotating it so that $\mathbf{r}_C = h(t)\,\mathbf{k}$ with the prescribed law $h(t) = H + b\sin\Omega t$ (H, b, Ω constants). A point particle P of mass m rests on the **outer** surface of the sphere, which is smooth, and can leave it. Use spherical coordinates about C: polar angle θ from the upward vertical and azimuth φ about Oz, with the basis $\{\mathbf{u}_r, \mathbf{e}_\theta, \mathbf{e}_\varphi\}$, so that $\mathbf{r}_P = h\,\mathbf{k} + R\,\mathbf{u}_r$ while P is in contact. Decide whether each statement is TRUE or FALSE and justify the answer. A verdict without a justification scores nothing.*

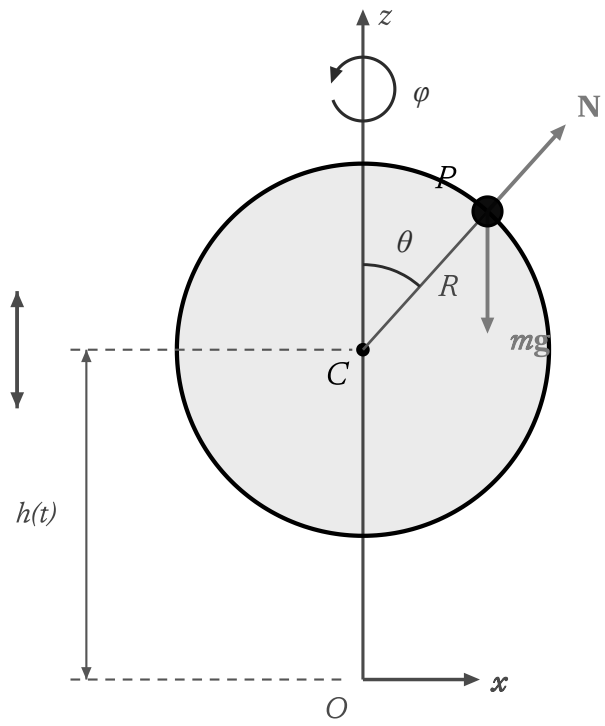


Figure 1. Sphere of radius R with centre C at the prescribed height $h(t) = H + b\sin\Omega t$ on the vertical axis; particle P on its outer surface, polar angle θ from the upward vertical, azimuth φ about the axis.

Part (a) · 1 point

Degrees of freedom

Statement: "While P is in contact with the sphere it has a single degree of freedom, because two conditions are imposed on it: it must lie on the sphere, and the sphere moves with the law $h(t)$."

Step 1. Coordinates and constraint equations

A free particle has 3 degrees of freedom (Notes §2.7). Contact with the sphere imposes one scalar equation on the position of P:

$$f(\mathbf{r},t) = \sqrt{x^2 + y^2 + \big(z - h(t)\big)^2} - R = 0$$

The motion of the sphere, $h(t)$, is a known function of time: it enters this equation but does not add another equation linking x, y, z. It removes no further degree of freedom.

Step 2. Count

$3 - 1 = 2$ degrees of freedom; θ and φ fix the position at each instant through $\mathbf{r}_P = h(t)\,\mathbf{k} + R\,\mathbf{u}_r(\theta,\varphi)$. The Lagrangian in (θ,φ) gives two equations of motion, each with its own second derivative (their mass matrix has determinant $m^2R^4\sin^2\theta \neq 0$ off the poles), which confirms two independent coordinates. The statement is FALSE.

KEY POINTS

- FALSE.
- One constraint on P: $f(\mathbf{r},t) = |\mathbf{r} - h(t)\mathbf{k}| - R = 0$. The law $h(t)$ is data, not a second constraint.
- $3 - 1 = 2$ degrees of freedom while in contact, coordinates (θ,φ); 3 after P leaves the sphere.

Part (b) · 1 point

Classification and work of the reaction

Statement: "The constraint is holonomic, rheonomic and unilateral, and although the sphere is smooth its normal reaction does work on P in S_0."

Step 1. Classification (Notes §2.7)

The constraint $f(\mathbf{r},t) = 0$ involves positions and time but no velocities: **holonomic**. It depends explicitly on t through $h(t)$, $\partial f/\partial t = -(z-h)\dot{h}/|\mathbf{r} - h\mathbf{k}|$, which is not identically zero (it vanishes only where $z = h$ or at the instants with $\dot{h} = 0$): **rheonomic**. P lies on the outside and can lose contact: the true condition is $f \geq 0$ with $N \geq 0$, an inequality: **unilateral**.

Step 2. Velocity of P

Differentiating $\mathbf{r}_P = h\,\mathbf{k} + R\,\mathbf{u}_r$ with $\dot{\mathbf{u}}_r = \dot\theta\,\mathbf{e}_\theta + \sin\theta\,\dot\varphi\,\mathbf{e}_\varphi$ (chain rule on $\mathbf{u}_r(\theta(t),\varphi(t))$):

$$\mathbf{v}_P = \dot h\,\mathbf{k} + R\big(\dot\theta\,\mathbf{e}_\theta + \sin\theta\,\dot\varphi\,\mathbf{e}_\varphi\big), \qquad \dot h = b\Omega\cos\Omega t$$

Step 3. Power of the reaction

Smooth means the reaction is normal to the surface: $\mathbf{N} = N\,\mathbf{u}_r$. With $\mathbf{u}_r\cdot\mathbf{e}_\theta = \mathbf{u}_r\cdot\mathbf{e}_\varphi = 0$ and $\mathbf{u}_r\cdot\mathbf{k} = \cos\theta$:

$$\dot W_N = \mathbf{N}\cdot\mathbf{v}_P = N\dot h\cos\theta = Nb\Omega\cos(\Omega t)\cos\theta$$

At $t = 0$ with P at the top ($\theta = 0$) it equals $Nb\Omega \neq 0$. A smooth reaction does no work only against a **fixed** surface (Notes §6.4); here the surface pushes P as it moves. In the frame translating with C the velocity of P is $\mathbf{v}_{rel}$, tangent to the sphere, and there $\mathbf{N}\cdot\mathbf{v}_{rel} = 0$. All the claims hold: TRUE.

KEY POINTS

- TRUE.
- Holonomic: f depends on position and time only; rheonomic: $\partial f/\partial t$ not identically zero; unilateral: $f \geq 0$ (outer surface, $N \geq 0$, P may leave).
- $\mathbf{v}_P = \dot h\,\mathbf{k} + R\big(\dot\theta\,\mathbf{e}_\theta + \sin\theta\,\dot\varphi\,\mathbf{e}_\varphi\big)$, $\mathbf{N}\cdot\mathbf{v}_P = N\dot h\cos\theta = Nb\Omega\cos\Omega t\cos\theta \neq 0$ in general.
- In the frame translating with C the reaction does no work ($\mathbf{u}_r\cdot\mathbf{v}_{rel} = 0$).

Interpretation and checks

- **Dimensional check:** $N\dot h$ is N m s^{-1} = W; f is a length.
- **Limits:** $b \to 0$ (fixed sphere) gives a scleronomic constraint and $\dot W_N = 0$: mechanical energy is then conserved. If P were a bead threaded on a fixed meridian, a second constraint would leave 1 degree of freedom.
- **Common mistakes in the exam:** (1) counting a prescribed motion as an extra constraint; (2) "smooth implies no work" for a moving surface; (3) calling the constraint bilateral when the particle is only resting on the outside.